

Week 8 -retrieval pipeline

RAG Pipeline Enterprise v2.0 • Testleaf

DATA PIPELINE

- Convert to JSON
- Embeddings & Store

SEARCH ENGINES

- Query Preprocessing
- Vector Search
- BM25 Search
- Hybrid Search
- Score Fusion

AI PROCESSING

- Enterprise AI Stack
MongoDB Atlas • Mistral AI • Groq

Convert to JSON

Upload and convert Excel files

1 Select File 2 Configure 3 Convert

File Upload
Select data type and configure your Excel file

Data Type
TC Test Cases

Select Excel File (Test Cases)
Accepts .xlsx and .xls files

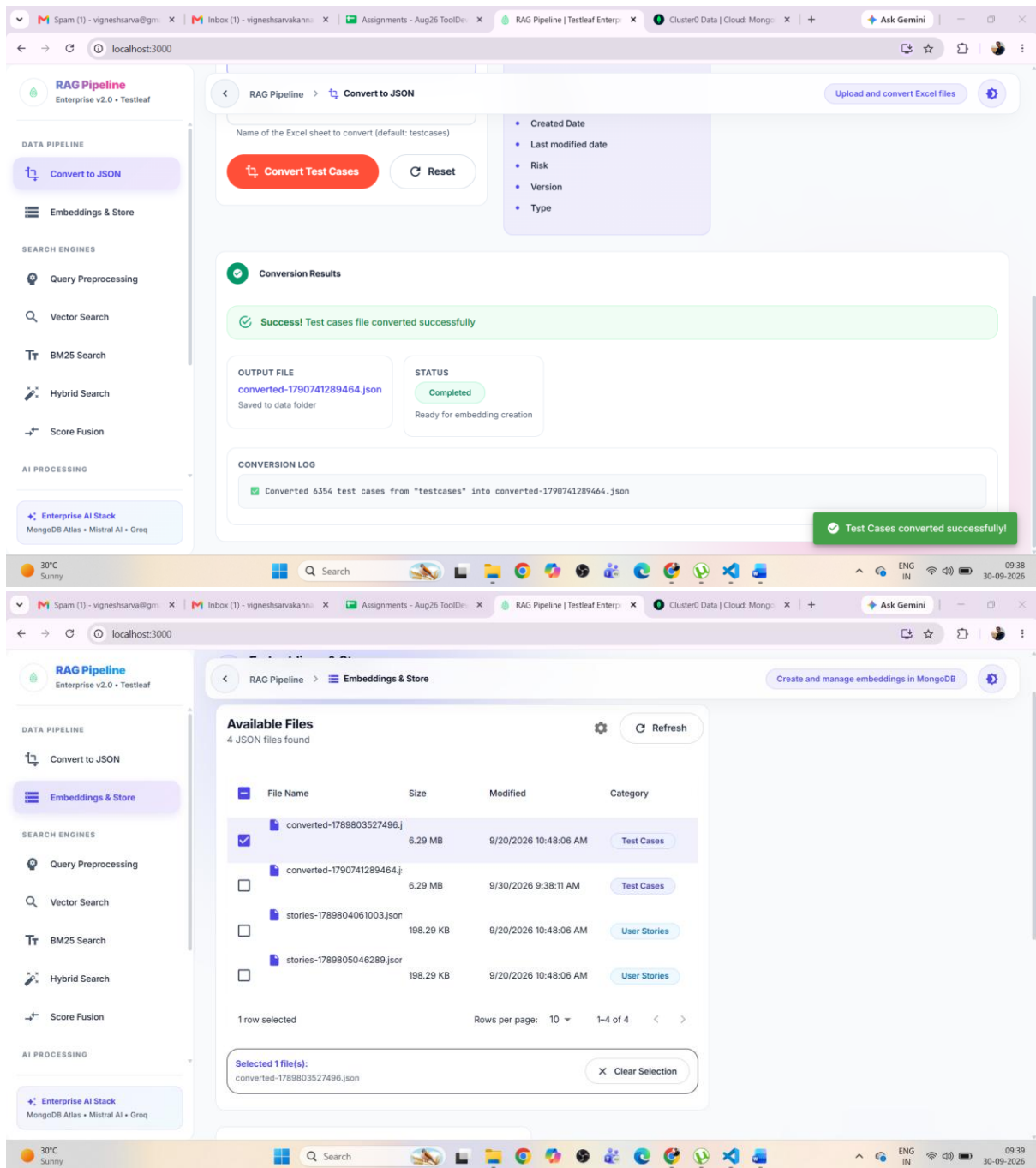
Sheet Name
testcases
Name of the Excel sheet to convert (default: testcases)

Convert Test Cases Reset

Expected Test Cases Columns
Your Excel file should contain these columns:

- Module
- Test ID
- Pre-Requisites
- Test Title
- Test Case Description
- Test Steps
- Expected Results
- Automation/Manual
- Priority
- Created By
- Created Date
- Last modified date
- Risk

File selected successfully



VS Code interface showing the development of a RAG Pipeline. The Explorer pane on the left displays the project structure, including files like `.env`, `src`, `scripts`, `search`, `rerank-search...`, `score-fusion...`, `search-comb...`, `search-jira-st...`, `search-vecto...`, `utilities`, `delete-all-do...`, `embeddings...`, `groqClient.js`, `llmService.js`, `mistralEmbe...`, `uploads`, `env`, `env.example`, `.gitignore`, `package-lock.json`, `package.json`, `Outline`, and `Timeline`.

The main editor displays the `.env` file with the following configuration:

```
.env
23 RERANK_PROVIDER="groq"
24 RERANK_MODEL="openai/gpt-oss-120b"
25
26 # 3. Summarization & Deduplication LLM (groq | gemini | openai |
27 SUMMARIZATION_PROVIDER="groq"
28 SUMMARIZATION_MODEL="qwen/qwen3.8-72b"
29
30 # ===== Token Optimization & Budget Control =====
31 # Enable token budgeting (NVIDIA models support 32k-128k context window)
32 ENABLE_TOKEN_OPTIMIZATION="true"
33 MAX_TOTAL_TOKENS="32000"
34 MAX_GENERATION_TOKENS="4000"
35 MAX_SUMMARY_TOKENS="800"
36
37 # ===== Automatic Fallback Mechanism =====
38 # Enable automatic fallback on rate limits (HTTP 429) or server outage
39 ENABLE_FALLBACK="true"
```

The Terminal pane shows the output of the application, indicating successful embedding and document insertion:

```
[0] [Batch 1/128] Embedding 50 items with Mistral AI...
[0] [Batch 1/128] Inserted 50 documents.
[0] [Batch 2/128] Embedding 50 items with Mistral AI...
[0] [Batch 2/128] Inserted 50 documents.
[0] [Batch 3/128] Embedding 50 items with Mistral AI...
[0] [Batch 3/128] Inserted 50 documents.
[0] [Batch 4/128] Embedding 50 items with Mistral AI...
[0] [Batch 4/128] Inserted 50 documents.
[0] [Batch 5/128] Embedding 50 items with Mistral AI...
```

The browser window displays the RAG Pipeline Enterprise v2.0 interface. The left sidebar shows the navigation menu with options like **DATA PIPELINE**, **Convert to JSON**, **Embeddings & Store**, **SEARCH ENGINES**, **Query Preprocessing**, **Vector Search**, **BM25 Search**, **Hybrid Search**, **Score Fusion**, and **AI PROCESSING**.

The main content area shows the **Embeddings & Store** section, displaying a table of embeddings and a progress bar for processing embeddings. The progress bar indicates 13% completion (850/6354 items).

The **Actions** section shows the process selected for the embeddings, with a target MongoDB collection set to `Auto-detect (test_cases)`.

Spam (1) - vigneshsava@gm... Inbox (1) - vigneshsava@... Assignments - Aug26 ToolDe... RAG Pipeline | Testleaf Enterp... Cluster0 Data | Cloud: Mongo... Ask Gemini

localhost:3000

RAG Pipeline

Enterprise v2.0 • Testleaf

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Embeddings & Store

Create and manage embeddings in MongoDB

READY TO EMBED

MONGODB COLLECTION: TEST_CASES

1 file(s) selected. Ready to store embeddings in MongoDB.

Vector Model: Mistral Embeddings (1024 dimensions)
Database: Stored directly into MongoDB Atlas collections

SELECTED FILES

converted-1789803527496.json

Processing Results

1 completed, 0 failed

converted-1789803527496.json completed

Processed 6354 Items: 6354 inserted, 0 failed

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rag-mongo-demo-E2E

Explorer

- Open Editors
- .env
- rag-mongo-demo-E2E
 - src
 - scripts
 - search
 - rerank-search... U
 - score-fusion... U
 - search-comb... U
 - search-jira-st... U
 - search-vecto... U
 - utilities
 - delete-all-do... U
 - embeddingS... U
 - groqClientjs U
 - llmServicejs U
 - mistralEmbe... U
 - uploads
 - .env
 - .env.example U
 - gitignore U
 - package-lockjson U
 - packagejson U
 - Outline
 - Timeline

```
.env
23 RERANK_PROVIDER="groq"
24 RERANK_MODEL="openai/gpt-oss-120b"
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33 MAX_TOTAL_TOKENS="32000"
34 MAX_GENERATION_TOKENS="4000"
35 MAX_SUMMARY_TOKENS="800"
36
37 # ===== Automatic Fallback Mechanism =====
38 # Enable automatic failover on rate limits (HTTP 429) or server outage
39 ENABLE_FALLBACK="true"
```

Problems Output Terminal

```
[0] [Batch 126/128] Embedding 50 items with Mistral AI...
[0] [Batch 126/128] Inserted 50 documents.
[0] [Batch 127/128] Embedding 50 items with Mistral AI...
[0] [Batch 127/128] Inserted 50 documents.
[0] [Batch 128/128] Embedding 4 items with Mistral AI...
[0] [Batch 128/128] Inserted 4 documents.
[0]
[0] [IN-PROCESS BATCH COMPLETE] 6354 items inserted, 0 failed, Total Cost: $0.132336
```

Chat

Sessions

Tip: Use /create-skill to create a skill the agent can load when relevant.

+ .env

Describe what to build

+ Agent Auto Balance

Local Default permissions

Ln 51, Col 70 Spaces: 4 UTF-8 LF {} Dotenv Autocomplete (0)

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Spam (1) - vigneshsarva@gm... x | Inbox (1) - vigneshsarvakann... x | Assignments - Aug26 ToolDe... x | RAG Pipeline | Testleaf Enterp... x | Cluster0 Data | Cloud: Mongo... x | + | Ask Gemini | - | x

cloud.mongodb.com/v2/6a9eb17927df8f29839faa4a#explorer/6a9eb1f601f2dba751092239/RAG_testcases/testcases/find

ORGANIZATION
vignesh's Org - 2026... PROJECT
Project 0

Data Explorer

- My Queries
- Data Modeling

CLUSTERS (1)

- Cluster0
 - RAO_testcases
 - testcases
 - userstories
 - admin
 - local
 - sample_mflix

Cluster0 > RAO_testcases > testcases

Documents 6.4K Aggregations Schema Indexes 1 Validation

Type a query: { field: 'value' } or [Generate query](#)

[EXPLAIN](#) [Reset](#) [Find](#) [Options](#)

[ADD DATA](#) [BULK](#) [EXPORT CODE](#)

100 1 - 100 of 6354

```
{
  "_id": ObjectId("6aa93a57826a5125ac6fae62"),
  "module": "Registration",
  "id": "IC_024",
  "preRequisites": "Valid UHID ",
  "title": "Smart Card",
  "description": "Verify user able to print the Smart Card",
  "steps": "1.Launch the application\n2.Login to the application with valid credentials.\n3.Click on Patient Service Tab and select Registration from the drop down...",
  "expectedResults": "Should be able to print Smart Card",
  "automationManual": "Automated",
  "priority": "P1",
  "createdBy": "Sumitha",
  "createdDate": 45064,
  "lastModifiedDate": 45087,
  "risk": "Low",
  "version": "v10.0",
  "type": "Interaction"
}
```

Alert System Status: All Good

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Spam (1) - vigneshsarva@gm... x | Inbox (1) - vigneshsarvakann... x | Assignments - Aug26 ToolDe... x | RAG Pipeline | Testleaf Enterp... x | Cluster0 Data | Cloud: Mongo... x | + | Ask Gemini | - | x

localhost:3000

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RAG Pipeline > Vector Search

Semantic AI vector search

Search Query

Enter your search terms and find similar test cases

Enter your search query

Share Diagnostic Reports with Patients via WhatsApp

Use descriptive terms related to test functionality

TRY AN EXAMPLE QUERY

- Share Diagnostic Reports with Patients via...
- Login test cases for the Billing module wi...
- Validate payment gateway timeout and retry...
- Password reset flow across web and mobile
- High-risk regression scenarios for checkou...

Advanced Metadata Filters [Show Filters](#)

Results Limit 5 Results

Search

Search Query: "Share Diagnostic Reports with Patients via WhatsApp"

Performance: \$0.000001 cost 14 tokens

5 results

Search Results

5 results ranked by semantic similarity

cloud.mongodb.com/v2/6a9eb17927df8f29839faa4a#explorer/6a9eb1f601f2dba751092239/RAG_testcases/testcases/find

ORGANIZATION: vignesh's Org - 2026-... PROJECT: Project 0

Data Explorer

- My Queries
- Data Modeling

CLUSTERS (1)

- Cluster0
 - RAO_testcases
 - testcases
 - userstories
 - admin
 - local
 - sample_mflix

Cluster0 > RAO_testcases > testcases

Documents 6.4K Aggregations Schema Indexes 1 Validation

Type a query: { field: 'value' } or [Generate query](#)

[ADD DATA](#) [BULK](#) [EXPORT CODE](#) 100 1 - 100 of 6354

```
{
  "_id": ObjectId("6aa93a57826a5125ac6fae62"),
  "module": "Registration",
  "id": "TC_024",
  "preRequisites": "Valid UHID ",
  "title": "Smart Card",
  "description": "Verify user able to print the Smart Card",
  "steps": "1.Launch the application\n2.Login to the application with valid credentials.\n3.Click on Patient Service Tab and select Registration from the drop down..."
  "expectedResults": "Should be able to print Smart Card",
  "automationManual": "Automated",
  "priority": "P1",
  "createdBy": "Sumitha",
  "createdDate": 45064,
  "lastModifiedDate": 45087,
  "risk": "Low",
  "version": "v10.0",
  "type": "Integration"
}
```

Alert System Status: All Good

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localhost:3000

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- [Enterprise AI Stack](#)
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RAG Pipeline > Vector Search

Semantic AI vector search

Search Results
5 results ranked by semantic similarity

Test Case	Similarity	Description	Source
CONSULTANT DASHBOARD - MY APPOINTMI	90.3%	Verify user able to Save Diagnosis	Doctors-Wards
REPORTS - CLINICAL REPORTS - VIEW REPO	90.2%	Verify user able to view Clinical Report	Doctors-Wards
REPORTS - CLINICAL REPORTS - VIEW REPO	90.1%	Verify user able to view Clinical Report	Doctors-Wards
DOCTOR DASHBOARD - OP DASHBOARD -CO	90.1%	Verify user able to Save Diagnosis	Doctors-Wards

Rows per page: 10 1-5 of 5

Detailed View

Consultant Dashboard - My Appointments - - OP Dashboard -Consultations - Diagnosis

Score: 0.9030 Doctors-Wards

Spam (1) - vigneshsarva@gm...

Inbox (1) - vigneshsarva@...

Assignments - Aug26 ToolDe...

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Cluster0 Data | Cloud: Mongo...

+

Ask Gemini

localhost:3000

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RAG Pipeline

BM25 Search

Keyword-based search algorithm

Fast keyword-based search using BM25 algorithm. Best for exact matches and term queries.

Search Query

Share Diagnostic Reports with

Results Limit

10

Search

Show Filters

BM25 Search Results: Found 10 test cases in 335ms

Search Type: bm25

Search Results (10)

#1

TC_2736

Score: 38.28

AHCSummary ->AHCSummary ->Diagnostic & Imaging report

Module: AHC

P3

Risk: Low

Automated

Description: Verify Diagnostic & Imaging report

Steps: 1.Launch the application 2.Login to the application with valid credentials. 3.Click on Patient Service Tab and select AHC from drop down. 4.Navigate to Functions -> AHCSummary ->AHCSummary. 5.Select date or Enter UHID/AHC No and click on "Go" button. 6.Click on AHC Number. 7.Click on Diagnostic & Imaging link 8.Observe the results 9.Click on Download or print icon

Expected Results: Diagnostic & Imaging details should be displayed

Created: 9/15/2026

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Search

ENG IN

09:40 30-09-2026

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RAG Pipeline

Hybrid Search

Combined BM25 + Vector search

Search Weights

BM25 Weight (Keyword): 65%

Vector Weight (Semantic): 35%

0% 50% 100%

0% 50% 100%

Balanced (50/50)

Keyword Heavy (70/30)

Semantic Heavy (30/70)

Show Filters

Hybrid Search Results: Found 10 test cases in 493ms

Stats: 5 in both indexes,5 BM25 only,0 vector only

Timing: BM25: 92ms, Vector: 395ms

Cost: \$0.000001 (14 tokens)

Search Results (10)

#1

TC_1174

Hybrid: 80.9%

Found in: both

Investigation report of patients

Module: IP digital

P2

Risk: Low

Automated

Description: Investigation report of patients

Found 10 results (5 in both indexes) in 493ms

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Search

ENG IN

09:40 30-09-2026

30°C Sunny 09:41 30-09-2026